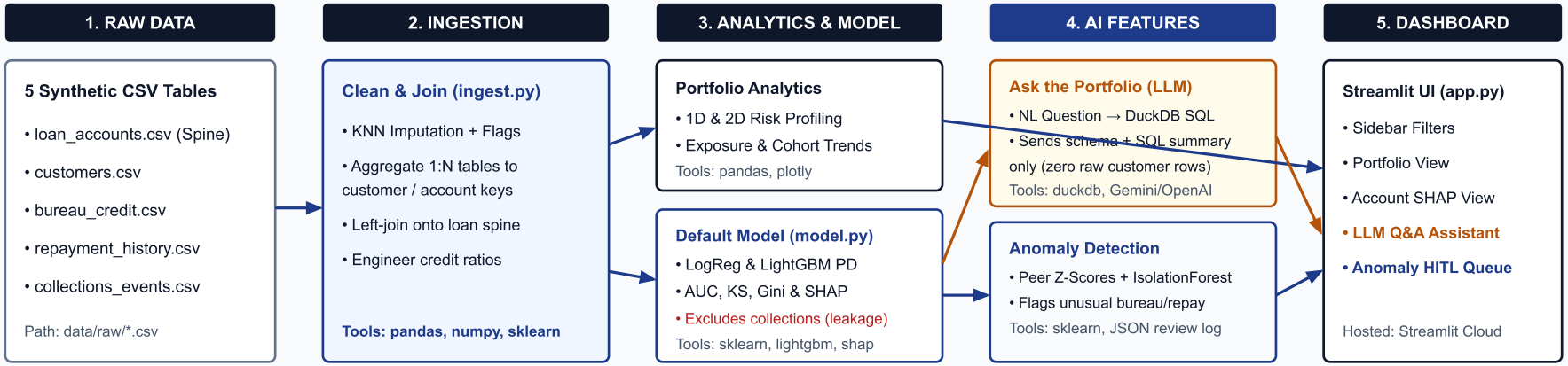
Credit Risk Portfolio Analytics — System & AI Architecture

End-to-End Flow: Raw Data → Ingestion → Analytics & Model → AI Features → Dashboard

Document: docs/architecture.pdf
Grain: Account-Level Master Dataset



STAGE OVERVIEW, TOOLS USED & LLM DATA BOUNDARY

Stage	Tools & Libraries	Core Responsibility	LLM Call & Data Sent
1. Ingestion (`src/ingest.py`)	`pandas`, `numpy`, `scikit-learn` (`KNNImputer`)	Cleans missing values, aggregates 1-to-many tables (bureau, repayments, collections) to parent keys, and left-joins them onto `loan_accounts`.	None (Local processing only).
2. Analytics (`src/analytics.py`)	`pandas`, `plotly`	Profiles portfolio risk across income, employment, region, age, and loan type, plus cross-dimensional cuts.	None (Local processing only).
3. Default Model (`src/model.py`)	`scikit-learn`, `lightgbm`, `scipy`, `shap`	Predicts `target_default` (excluding collections leakage) using Logistic Regression and LightGBM; outputs PD, AUC, KS, Gini, and SHAP drivers.	None (Local processing only).
4. AI: Ask the Portfolio (`src/ai_assistant.py`)	`duckdb`, `google-genai` / `openai`	Translates natural-language questions into DuckDB SQL over the master dataset and returns grounded answers.	**Called Here:** Sends only table schema metadata + user question to generate SQL, then sends the small aggregated SQL result table to format the final answer.
5. AI: Anomaly Detection (`src/anomaly.py`)	`scikit-learn` (`IsolationForest`), `streamlit`	Flags unusual bureau and repayment patterns vs. peer segments and logs analyst Confirm/Dismiss decisions.	None (Runs locally; persists analyst decisions to JSON).